

Table 1: The IoT journey

	Key points	Details/Examples
Growth and industry impact	Evolution from simple connected devices to a large ecosystem	Initially applied in asset tracking and remote monitoring; now central to healthcare, manufacturing, transportation, and smart cities
	Operational efficiency and innovation	IoT enables better processes, predictive maintenance, and improved decision-making
	Career opportunities	New roles in technology, analytics, network management, IoT solution development, and operations
Role of networks and data	Connectivity as the foundation	Reliable network connections allow seamless communication between devices
	Data collection, processing, and analysis	Turning raw data into actionable insights for decision-making and process optimisation
	Scaling IoT with carriers and platforms	Cloud platforms, carriers, and data management systems help IoT expand efficiently
	Interplay between devices, networks, and data	Understanding this interaction explains IoT growth and its industry impact

to create smarter, efficient systems, and we have coined the acronym AIoT to refer to this partnership. In this relationship, IoT provides the data from connected devices, and AI analyses it to create insights, improve operations, and enable prediction-based actions. The combination of IoT with AI offers the opportunity to use them across industries, from smart homes and healthcare to industrial automation and logistics.

Overlap and rivalry myths: IoT and AI rarely directly compete with one another, despite the common assumption that they do. AI adds intelligence and decision-making to the connectivity and data collection that IoT provides. Any suggestion of competition is a misinterpretation of one technology as an improvement over another. Both complement one another, which opens more possibilities for innovation and value creation.

Collaborative potential: A self-learning and adaptable ecosystem will be produced if artificial intelligence (AI) and the Internet of Things (IoT) are successfully integrated. Predictions, resource optimisation, and real-time change response will all be possible for businesses. In addition to opening new career paths in fields as diverse as data analytics, IoT engineering, and smart system management, this will improve how networks and devices function.

Table 2: The AI impact on IoT

	Key points	Details/Examples
Differences in scope	Complementary roles of AI and IoT	IoT connects devices and gathers data; AI analyses, interprets, and acts on that data
	Bringing intelligence to IoT	Transforms raw device data into predictive insights, automation, and smarter decision-making
	Expanding beyond connectivity	AI adds new layers of productivity and innovation to IoT applications
AI adoption and dominance	Rapid incorporation into IoT systems	AI is being integrated across industries: manufacturing (predictive maintenance), smart cities (traffic management), healthcare, etc
	Extending IoT capabilities	IoT becomes a learning, adaptive, and self-optimising ecosystem
	Changing engagement with technology	Businesses and professionals interact with IoT in more intelligent, automated, and efficient ways